$P_3(\lambda) = (a_3-\lambda)P_2(\lambda) - b_3^2 P_1(\lambda)$

$= (1.9232-\lambda)(\lambda^2 - 1.0768\lambda - 13) - (-0.3844)^2(1-\lambda)$

$= 1.9232\lambda^2 - 2.0709\lambda - 25.0016 - \lambda^3 + 1.0768\lambda^2 + 13\lambda - 0.14(1-\lambda)$

$= -\lambda^3 + 3\lambda^2 + 11\lambda - 25$

$= -\lambda^3 + 3\lambda^2 + 11.1478\lambda - 25.1478$

$\lambda_1 = -3.12,\ 4.2299,\ 1.8997$

## module-5

## Picard's method

$\frac{dy}{dx} = f(x,y),\ y(x_0) = y_0$

$dy = f(x,y).dx$

In LHS w.r.t $y$ by taking limits as $y_0$ to $y$

" " as $x_0$ to $x$

RSH

$\int_{y_0}^{y} dy = \int_{x_0}^{x} f(x,y)\,dx$

$y]_{y_0}^{y} = \int_{x_0}^{x} f(x,y)\,dx$

$y - y_0 = \int_{x_0}^{x} f(x,y)\,dx$

$y = y_0 + \int_{x_0}^{x} f(x,y)\,dx \rightarrow (1)$

I^st^ app^n^ put $y = y_n$ in eq^n^ (1)

$y_1 = y_0 + \int_{x_0}^{x} f(x,y_0)\,dx$

$II^{nd}$ $app^n$ put $y=y_1$ in $eq^n$ ①

$$y_2 = y_0 + \int_{x_0}^{x} f(x, y_1)\,dx$$

III $app^n$

$$y_3 = y_0 + \int_{x_0}^{x} f(x, y_2)\,dx$$

1) Use Picards method to obtain 4 approx to find the sol of $\frac{dy}{dx} = x+y$, $y(0)=1$ & hence find $y$ at $x = 0.1$ & $0.2$

$y_0 = 1$, $x_0 = 0$

$dy = (x+y)\,dx$

$$\int_{y_0}^{y} dy = \int_{x_0}^{x} (x+y)\,dx$$

$$\int_{1}^{y} dy = \int_{0}^{x} (x+y)\,dx$$

$$y\Big]_1^y = \cancel{0} \int_{0}^{x} (x+y)\,dx$$

$$y-1 = \cancel{1+} \int_{0}^{x} (x+y)\,dx$$

$$y = 1 + \int_{0}^{x} (x+y)\,dx$$

$I^{st}$ $app^n$

$$y_1 = 1 + \int_{0}^{x} (x+1)\,dx$$

$$y_1 = 1 + \left[\frac{x^2}{2} + x\right]_0^x$$

$$y_1 = 1 + x + \frac{x^2}{2}$$

$II^{nd}$ $app^n$

$$y_2 = 1 + \int_{0}^{x} \left(x + 1 + x + \frac{x^2}{2}\right) dx$$

$$y_2 = 1 + \int_{0}^{x} \left(1 + 2x + \frac{x^2}{2}\right) dx$$

$$= 1 + \left[x + x^2 + \frac{x^3}{6}\right]_0^x$$

$$= 1 + x + x^2 + \frac{x^3}{6}$$

III $app^n$

$$y_3 = 1 + \int_{0}^{x} \left(x + 1 + x + x^2 + \frac{x^3}{6}\right) dx$$

$$= 1 + \int_{0}^{x} \left(1 + 2x + x^2 + \frac{x^3}{6}\right) dx$$

$$= 1 + \left[x + x^2 + \frac{x^3}{3} + \frac{x^4}{24}\right]_0^x$$

$$y_3 = \left(1 + x + x^2 + \frac{x^3}{3} + \frac{x^4}{24}\right]$$

IV appn

$$y_4 = 1 + \int_0^x \left(x + 1 + x + x^2 + \frac{x^3}{3} + \frac{x^4}{24}\right) dx$$

$$= 1 + \int_0^x \left(1 + 2x + x^2 + \frac{x^3}{3} + \frac{x^4}{24}\right) dx$$

$$= 1 + \int_0^x \left[x + \frac{x^2}{\cancel{2}} + \frac{x^3}{3} + \frac{x^4}{12} + \frac{x^5}{120}\right]_0^x$$

$$y_4 = 1 + x + x^2 + \frac{x^3}{3} + \frac{x^4}{12} + \frac{x^5}{120}$$

$x = 0.1 \longrightarrow 1 + 0.1 + (0.1)^2 + \frac{(0.1)^3}{3} + \frac{(0.1)^4}{12} + \frac{(0.1)^5}{120} = 1.1103$

$x = 0.2 \longrightarrow 1 + 0.2 + (0.2)^2 + \frac{(0.2)^3}{3} + \frac{(0.2)^4}{12} + \frac{(0.2)^5}{120} = 1.2428$

2) $\frac{dy}{dx} = y\cancel{x} + e^x$ $\qquad y(0) = 1$

$$y = 1 + \int_0^x (y + e^x) dx$$

$dy = (\cancel{x}y + e^x) dx$

$\int_{y_0}^{y} dy = \int_0^x (y + e^x) dx$

$\int_1^y dy = \int_0^x (y + e^x) dx$

$[y]_1^y = \int_0^x (y + e^x) dx$

$[y - 1] = \int_0^x (y + e^x) dx$

Ist appn

$$y_1 = 1 + \int_0^x (1 + e^x) dx$$

$$= 1 + [x + e^x]_0^x$$

$$= \cancel{1} + x + e^x \cancel{-1}$$

$$\underline{y_1 = x + e^x}$$

$y_2 = 1 + \int_0^x (y + x + e^x)\, dx$

II$^{nd}$ app

$y_2 = 1 + \int_0^x [x + e^x + e^x$

$y_2 = 1 + \int_0^x (x + 2e^x)\, dx$

$= 1 + \left[\frac{x^2}{2} + 2e^x\right]_0^x$

$y_2 = 1 + \frac{x^2}{2} + 2e^x - 2$

$y_2 = \frac{x^2}{2} + 2e^x - 1$

III$^{rd}$

$y_3 = 1 + \int_0^x \left(\frac{x^2}{2} + 2e^x - 1 + e^x\right) dx$

$= 1 + \int_0^x \left(\frac{x^2}{2} + 3e^x - 1\right) dx$

$= 1 + \left[\frac{x^3}{6} + 3e^x - x\right]_0^x$

$y_3 = 1 + \frac{x^3}{6} + 3e^x - x - 3 = \frac{x^3}{6} + 3e^x - x - 2$

IV app$^n$

$y_4 = 1 + \int_0^x \left(\frac{x^3}{6} + 3e^x - x - 2 + e^x\right) dx$

$= 1 + \int_0^x \frac{x^3}{6} + 4e^x - x - 2)\, dx$

$= 1 + \left[\frac{x^4}{24} + 4e^x - \frac{x^2}{2} - 2x\right]_0^x$

$= 1 + \frac{x^4}{24} + 4e^x - \frac{x^2}{2} - 2x - 4$

$y_4 = \frac{x^4}{24} + 4e^x - \frac{x^2}{2} - 2x - 3$

3) $x = 0.1$ given that $y_0 = 1$ when $x_0 = 0$ & the diff eq$^n$ is

$\frac{dy}{dx} = 3x + y^2$

$dy = (3x + y^2)\, dx$

$\int_{y_0}^{y} dy = \int_0^x (3x + y^2)\, dx$

$\int_1^y dy = \int_0^x (3x + y)^2\, dx$

$[y]_1^y = \int_0^x (3x+y^2)\,dx$

$y-1 = \int_0^x (3x+1)\,dx$

I st app

$y_1 = 1+\int_0^x (3x+1)\,dx$

$= 1+\left(\frac{3x^2}{2}+x\right)_0^x$

$= 1+\frac{3x^2}{2}+x$

II nd app$^n$ $a^2+b^2+c^2+2ab+2bc+2ca$

$y_2 = 1+\int_0^x \left(3x+\left(1+\frac{3x^2}{2}+x\right)^2\right)dx$

$\cancel{= 1+\left[\frac{3x^2}{2}+x+3x^3\right.}$

$= 1+\int_0^x \left(3x+1+\frac{9x^4}{4}+x^2+3x^2+3x^3+2x\right)dx$

$= 1+\int_0^x \left(1+\frac{9x^4}{4}+3x^3+4x^2+5x\right)dx$

$\cancel{= 1+\frac{9x^4}{4}+3x^3+4x^2+5x}$

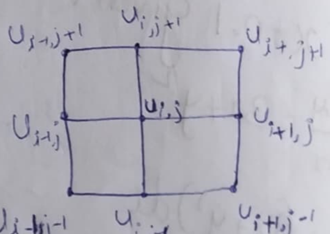
$= 1+\left[x+\frac{9x^5}{20}+\frac{3x^4}{4}+\frac{4x^3}{3}+\frac{5x^2}{2}\right]_0^x$

$= 1+x+\frac{9x^5}{20}+\frac{3x^4}{4}+\frac{4x^3}{3}+\frac{5x^2}{2}$

## Laplace

$\nabla^2\phi = 0$

$\frac{\partial^2\phi}{\partial x^2}+\frac{\partial^2\phi}{\partial y^2} = 0$




$u_{ij} = \frac{1}{4}\left[u_{i,j+1}+u_{i,j-1}+u_{i-1,j}+u_{i+1,j}\right]$ → Standard five point formula

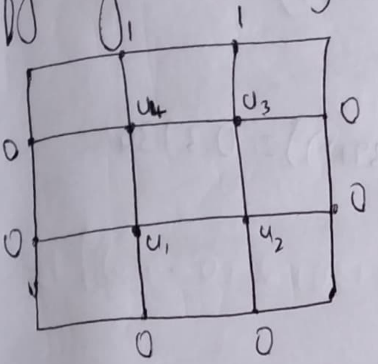
$u_{ij} = \frac{1}{4}\left[u_{i+1,j+1}+u_{i+1,j-1}+u_{i-1,j+1}+u_{i-1,j-1}\right]$ → Standard diagnol formula.

# SOR (Successive Over Relaxation)

$$u_{ij}^{n+1} = u_{ij}^{n} + \frac{\omega}{4}\left[u_{i-1,j}^{n+1} + u_{i+1,j}^{n} + u_{i,j-1}^{n+1} + u_{i,j+1}^{n} - 4u_{ij}^{n}\right]$$

where $\omega = 1.1$

1) Solve the eqn $u(x,y)$ $u_{xx} + u_{yy} = 0$ in the domain of the figure given below by using SOR method Considering $\omega = 1.1$




$I^{st}$ iteration

SOR

$$u_{ij}^{n+1} = u_{ij}^{n} + \frac{\omega}{4}\left[u_{i-1,j}^{n+1} + u_{i+1,j}^{n} + u_{i,j-1}^{n+1} + u_{i,j+1}^{n} - 4u_{ij}^{n}\right]$$

$$u_1 = 0 + \frac{1.1}{4}\left[0+0+0+1-4(0)\right]$$

$$= 0.275$$

$$u_2 = 0 + \frac{1.1}{4}\left[0.275 + 0 + 0 + 1 - 4(0)\right] = 0.3506$$

$$u_3 = 0 + \frac{1.1}{4}\left[\cancel{0.275 + 0.3506}\ 0 + 0 + 0.3506 + 1 - 4(0)\right] = 0.3714$$

$$u_4 = 0 + \frac{1.1}{4}\left[0 + 0.3714 + 0.275 + 1 - 4(0)\right] = 0.4528$$

$II^{nd}$ iteration

$$u_1 = 0.275 + \frac{1.1}{4}\left[0 + 0 + 0.3506 + 0.4528 - 4(0.275)\right] = 0.1934$$

$$u_2 = 0.3506 + \frac{1.1}{4}\left[0 + 0 + \overset{0.1934}{\cancel{0.275}} + 0.3714 - 4(0.3506)\right] = \cancel{0.1427}\ 0.1203$$

$$u_3 = 0.3714 + \frac{1.1}{4}\left[0 + 1 + 0.1203 + 0.4528 - 4(0.3714)\right] = 0.3955$$

$$u_4 = 0.4528 + \frac{1.1}{4}\left[1 + 0 + 0.1934 + 0.3955 - 4(0.4528)\right] = 0.3917$$

III$^{rd}$ iteration

$$u_1 = 0.1934 + \frac{1.1}{4}\left[0+0+0.1203+0.3917-4(0.1934)\right] = 0.1215$$

$$u_2 = 0.1203 + \frac{1.1}{4}\left[0+0+\overset{0.1215}{\cancel{0.1934}}+0.3955-4(0.1203)\right] = 0.1301$$

$$u_3 = 0.3955 + \frac{1.1}{4}\left[0+1+0.1301+0.3917-4(0.3955)\right] = 0.3789$$

$$u_4 = 0.3917 + \frac{1.1}{4}\left[1+0+0.1215+0.3789-4(0.3917)\right] = 0.3734$$

IV iteration

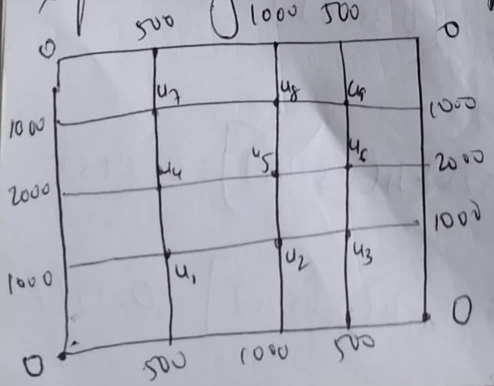
$$u_1 = 0.1215 + \frac{1.1}{4}\left[0+0+0.1301+0.3734-4(0.1215)\right] = 0.1263$$

$$u_2 = 0.1301 + \frac{1.1}{4}\left[0+0+0.1263+0.3789-4(0.1301)\right] = 0.1259$$

$$u_3 = 0.3789 + \frac{1.1}{4}\left[0+1+0.1259+0.3734-4(0.3789)\right] = 0.3744$$

$$u_4 = 0.3734 + \frac{1.1}{4}\left[1+0+0.1263+0.3744-4(0.3734)\right] = 0.3754$$

2) Solve $u_{xx}+u_{yy}=0$ using SOR method in the following square region with the boundary condition as indicated in fig.




Initial value

$$u_5 = \frac{1}{4}\left[1000+1000+2000+2000\right] = 1500$$

$$u_1 = \frac{1}{4}\left[0+1000+2000+1500\right] = 1125$$

$$u_3 = \frac{1}{4}\left[0+1000+2000+1500\right] = 1125$$

Diagonal

$u_7 = \frac{1}{4}[0+1000+2000+1500] = 1125$ } Diagonal

$u_9 = \frac{1}{4}[0+1000+2000+1500] = 1125$

$u_2 = \frac{1}{4}[1125+1000+1125+1500] = 1187.5$

$u_4 = \frac{1}{4}[2000+1125+1125+1500] = \cancel{1187.5}\ 1437.5$

$u_6 = \frac{1}{4}[1125+\overset{1500}{\cancel{1125}}+2000+1125] = \cancel{13}\ 1437.5$

$u_8 = \frac{1}{4}[1000+1125+1500+1125] = 1187.5$

SOR

$$u_{ij}^{n+1} = u_{ij}^{n} + \frac{\omega}{4}\left[u_{i-1\,j}^{n+1} + u_{i+1\,j}^{n} + u_{i\,j-1}^{n+1} + u_{i\,j+1}^{n+1} - 4u_{ij}^{n}\right]$$

$u_1 = 1125 + \frac{1.1}{4}[1000+500+1437.5+1187.5-4(1125)] = 1021.8\cancel{75}9$

$u_2 = 1187.5 + \frac{1.1}{4}[1021.\overset{9}{\cancel{875}}+1000+1500+1125-4(1187.5)] = 1159.1\cancel{375}$

$u_3 = 1125 + \frac{1.1}{4}[1159.1\cancel{375}+500+1000+1437.5-4(1125)] = 1014.1$

$u_4 = 1437.5 + \frac{1.1}{4}[2000+1021.9+1125+1500-4(1437.5)] = 1409.1$

$u_5 = 1500 + \frac{1.1}{4}[1409.1+1159.1+1187.5+1437.5-4(1500)] = 1278.1$

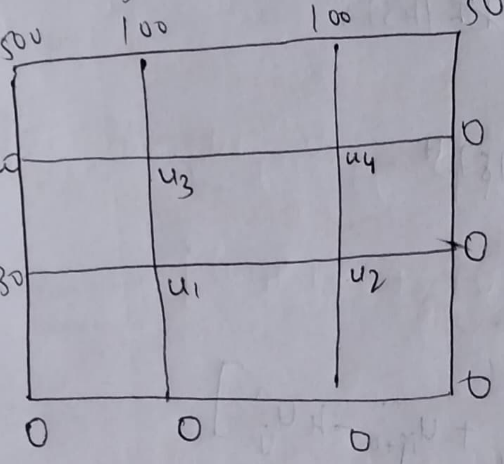
$u_6 = 1437.5 + \frac{1.1}{4}[1278.1+1014.1+1125+2000-4(1437.5)] = 1345.9$

$u_7 = 1125 + \frac{1.1}{4}[1000+500+1409.1+1187.5-4(1125)] = 1014.\cancel{0}061$

$u_8 = 1187.5 + \frac{1.1}{4}\left[1000 + 1278.1 + 1014.1 + 1125 - 4(1187.5)\right] = 1095.9$

$u_9 = 1125 + \frac{1.1}{4}\left[1095.9 + 500 + 1000 + 1345.9 - 4(1125)\right] = 971.49$

Solve $u_{xx} + u_{yy} = 0$, in the following square region with the boundary condition as indicated in fig




Ist iteration

$u_{ij}^{n+1} = u_{ij}^{n} + \frac{w}{4}\left[u_{i-1,j}^{n+1} + u_{i+1,j}^{n} + u_{i,j-1}^{n+1} + u_{i,j+1}^{n} - 4u\right.$

$u_1 = 0,\ u_2 = 0,\ u_3 = 20,\ u_4 = 0$

$u_1 = 0 + \frac{1.1}{4}\left[0 + 30 + 20 + 0 - 4(0)\right] = 13.75$

$u_2 = 0 + \frac{1.1}{4}\left[13.75 + 0 + 0 + 0 - 4(0)\right] = 3.7813$

$u_3 = 20 + \frac{1.1}{4}\left[13.75 + 20 + 100 + 0 - 4(20)\right] = 34.78$

$u_4 = 0 + \frac{1.1}{4}\left[3.7813 + 0 + 34.7813 + 100 - 4(0)\right]$

$= 38.1047$

IInd iteration

$u_1 = 13.75 + \frac{1.1}{4}\left[0 + 30 + 3.7813 + 34.7813 - 4(13.75)\right] = 17.47$

$u_2 = 3.7813 + \frac{1.1}{4}\left[0 + 0 + \overset{17.47}{\cancel{13.75}} + 38.1047 - 4(3.7813)\right] = 14.90$

$u_3 = 34.7813 + \frac{1.1}{4}\left[17.47 + 20 + 100 + 38.1047 - 4(34.7813)\right] = 44.80$

$u_4 = 38.1047 + \frac{1.1}{4}\left[0 + 100 + 14.90 + 44.80 - 4(38.1047)\right] = 40.10$

II$^{nd}$

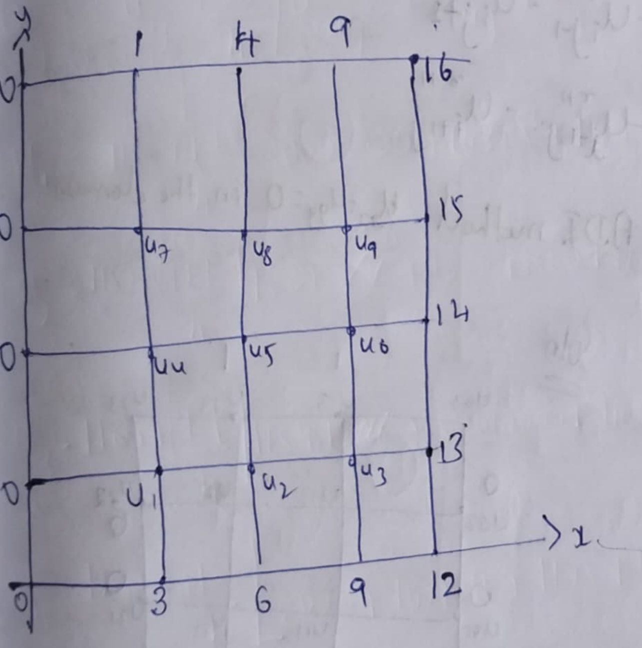
$u_1 = 17.47 + \frac{1.1}{4}\left[0+30+14.90+44.80-4(17.47)\right] = 22.92$

$u_2 = 14.90 + \frac{1.1}{4}\left[0+0+22.92+40.10-4(14.9)\right] = 15.84$

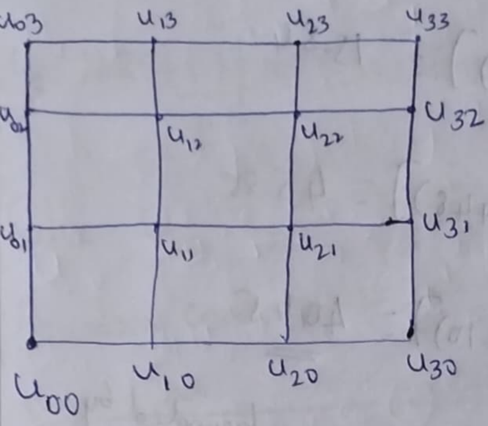
$u_3 = 44.8 + \frac{1.1}{4}\left[100+20+22.92+40.10-4(44.8)\right] = 45.85$

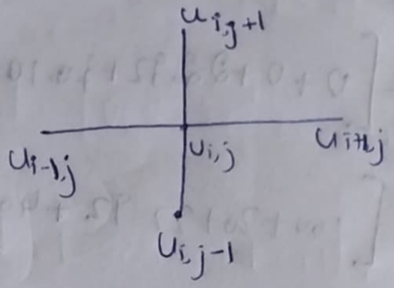
$u_4 = 40.10 + \frac{1.1}{4}\left[0+100+15.84+45.85-4(40.10)\right] = \underline{\underline{40.45}}$

Solve $u_{xx}+u_{yy}=0$ using SOR method in the square region bounded by coordinate axis & the lines $x=4$, $y=4$ with the boundary condition $u(0,y)=0$, for $0\le y\le 4$ & $u(4,y)=12+y$ for $0\le y\le 4$ and $u(x,0)=3x$ for $0\le x\le 4$ & $u(x,4)=x^2$ for $0\le x\le 4$.



# ADI (Alternative directed implicit) method:-




$$u_{ij} = \frac{1}{4}\left[u_{i-1,j} + u_{i+1,j} + u_{i,j+1} + u_{i,j-1}\right]$$

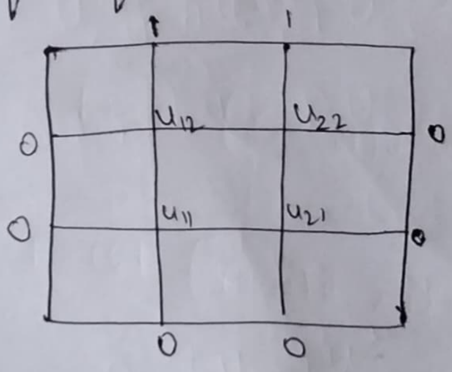
$$u_{i-1,j} + u_{i+1,j} + u_{i,j+1} + u_{i,j-1} - 4u_{ij} = 0$$

$$\star \quad u_{i-1,j} - 4u_{ij} + u_{i+1,j} = -u_{i,j-1} - u_{i,j+1}$$

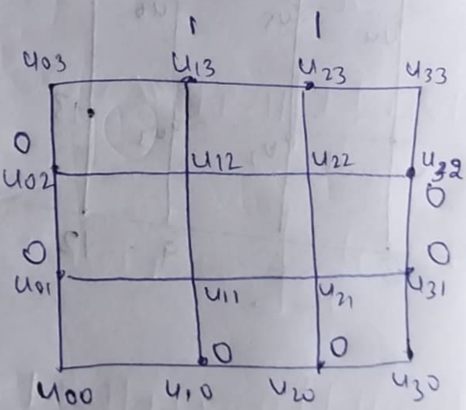
$$\star \quad u_{i,j-1} - 4u_{ij} + u_{i,j+1} = -u_{i-1,j} - u_{i+1,j}$$

$$\therefore u_{i-1,j}^{r+1} - 4u_{ij}^{r+1} + u_{i+1,j}^{r+1} = -u_{i,j-1}^{r} - u_{i,j+1}^{r}$$
&
$$u_{i,j-1}^{r+2} - 4u_{ij}^{r+2} + u_{i,j+1}^{r+2} = -u_{i-1,j}^{r+1} - u_{i+1,j}^{r+1}$$

1) Solve Laplace eqn using ADI method $u_{xx}+u_{yy}=0$, in the domain of the figure.




Soln




The ADI formula are given by $u_{i-1,j}^{r+1} - 4u_{ij}^{r+1} + u_{i+1,j}^{r+1} = -u_{i,j-1}^{r} - u_{i,j+1}^{r}$

$$u_{i,j-1}^{r+2} - 4u_{ij}^{r+2} + u_{i,j+1}^{r+2} = -u_{i-1,j}^{r+1} - u_{i+1,j}^{r+1}$$

In eq$^n$ (1) put $r=0$ & $j=1$

$$u^{(1)}_{i-1,1} - 4u^{(1)}_{i,1} + u^{(1)}_{i+1,1} = -u^{0}_{i,0} - u^{(0)}_{i,2} \rightarrow (3)$$

In eq$^n$ (3) put $i=1$ & $i=2$

$$\left.\begin{array}{l} u'_{0,1} - 4u'_{1,1} + u'_{2,1} = -u^{0}_{1,0} - u^{0}_{1,2} \\ u'_{1,1} - 4u'_{2,1} + u'_{3,1} = -u^{0}_{2,0} - u^{0}_{2,2} \end{array}\right\} \rightarrow (4)$$

Substitute values in eq$^n$ 4

$$0 - 4u'_{1,1} + u'_{2,1} = -0 - 1$$

$$u'_{1,1} - 4u'_{2,1} + 0 = -0 - 1$$

$$\underline{\underline{u'_{1,1} = 1/3}} \quad \underline{\underline{u'_{2,1} = 1/3}}$$

In eq$^n$ (1) $r=0$ & $j=2$

$$u'_{i-1,2} - 4u'_{i,2} + u'_{i+1,2} = -u^{0}_{i,1} - u^{0}_{i,3} \rightarrow (5)$$

In eq$^n$ (5) put $i=1$ & $i=2$

$$\left.\begin{array}{l} u'_{0,2} - 4u'_{1,2} + u'_{2,2} = -u^{0}_{1,1} - u^{0}_{1,3} \\ u'_{1,2} - 4u'_{2,2} + u'_{3,2} = -u^{0}_{2,1} - u^{0}_{2,3} \end{array}\right\} \rightarrow (6)$$

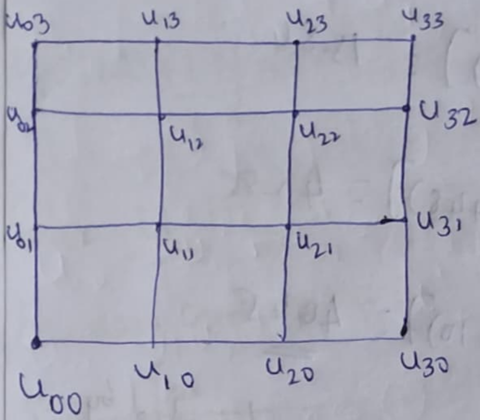
Substitute values in eq$^n$ (6)

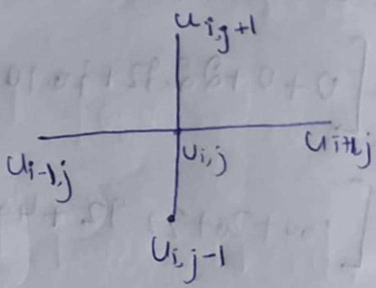
$$0 - 4u'_{1,2} + u'_{2,2} = -0 - 1$$

$$u'_{1,2} - 4u'_{2,2} + 0 = -0 - 1$$

$$\underline{\underline{u'_{1,2} = 1/3}} \quad \underline{\underline{u'_{2,2} = 1/3}}$$

# ADI (Alternative directed implicit) method:-




$$u_{ij} = \frac{1}{4}\left[u_{i-1,j} + u_{i+1,j} + u_{i,j+1} + u_{i,j-1}\right]$$

$$u_{i-1,j} + u_{i+1,j} + u_{i,j+1} + u_{i,j-1} - 4u_{ij} = 0$$

☆ $u_{i-1,j} - 4u_{ij} + u_{i+1,j} = -u_{i,j-1} - u_{ij+1}$

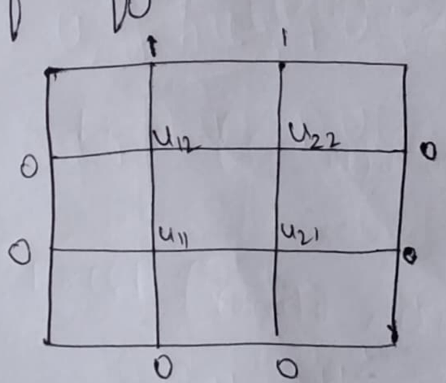
☆ $u_{i,j-1} - 4u_{ij} + u_{i,j+1} = -u_{i-1,j} - u_{i+1,j}$

$$\therefore u_{i-1,j}^{r+1} - 4u_{ij}^{r+1} + u_{i+1,j}^{r+1} = -u_{i,j-1}^{r} - u_{i,j+1}^{r}$$

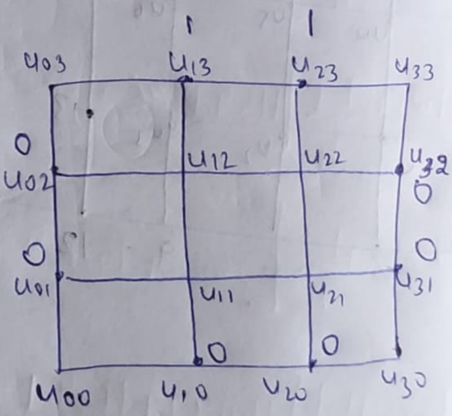
&

$$u_{i,j-1}^{r+2} - 4u_{ij}^{r+2} + u_{i,j+1}^{r+2} = -u_{i-1,j}^{r+1} - u_{i+1,j}^{r+1}$$

1) Solve Laplace eqn using ADI method $u_{xx}+u_{yy}=0$, in the domain of the figure.




Soln




The ADI formula are given by $u_{i-1,j}^{r+1} - 4u_{ij}^{r+1} + u_{i+1,j}^{r+1} = -u_{i,j-1}^{r} - u_{i,j+1}^{r}$

$$u_{i,j-1}^{r+2} - 4u_{ij}^{r+2} + u_{i,j+1}^{r+2} = -u_{i-1,j}^{r+1} - u_{i+1,j}^{r+1}$$

In eqⁿ ① put $r=0$ & $j=1$

$$u^{(1)}_{i-1,1} - 4u^{(1)}_{i,1} + u^{(1)}_{i+1,1} = -u^{0}_{i,0} - u^{(0)}_{i,2} \rightarrow ③$$

In eqⁿ ③ put $i=1$ & $i=2$

$$\left.\begin{array}{l} u^{1}_{0,1} - 4u^{1}_{1,1} + u^{1}_{2,1} = -u^{0}_{1,0} - u^{0}_{1,2} \\ u^{1}_{1,1} - 4u^{1}_{2,1} + u^{1}_{3,1} = -u^{0}_{2,0} - u^{0}_{2,2} \end{array}\right\} \rightarrow ④$$

Substitute values in eqⁿ 4

$$0 - 4u^{1}_{1,1} + u^{1}_{2,1} = -0 - 1$$

$$u^{1}_{1,1} - 4u^{1}_{2,1} + 0 = -0 - 1$$

$$\underline{\underline{u^{1}_{1,1} = 1/3}} \qquad \underline{\underline{u^{1}_{2,1} = 1/3}}$$

In eqⁿ ① $r=0$ & $j=2$

$$u^{1}_{i-1,2} - 4u^{1}_{i,2} + u^{1}_{i+1,2} = -u^{0}_{i,1} - u^{0}_{i,3} \rightarrow ⑤$$

In eqⁿ ⑤ put $i=1$ & $i=2$

$$\left.\begin{array}{l} u^{1}_{0,2} - 4u^{1}_{1,2} + u^{1}_{2,2} = -u^{0}_{1,1} - u^{0}_{1,3} \\ u^{1}_{1,2} - 4u^{1}_{2,2} + u^{1}_{3,2} = -u^{0}_{2,1} - u^{0}_{2,3} \end{array}\right\} ⑥$$

Substitute values in eqⁿ ⑥

$$0 - 4u^{1}_{1,2} + u^{1}_{2,2} = -0 - 1$$

$$u^{1}_{1,2} - 4u^{1}_{2,2} + 0 = -0 - 1$$

$$\underline{\underline{u^{1}_{1,2} = 1/3}} \qquad \underline{\underline{u^{1}_{2,2} = 1/3}}$$

In eq$^n$ ② put $r=0,\ i=1$

$$u^2_{1,j-1} - 4u^2_{1,j} + u^2_{1,j+1} = -u^1_{0,j} - u^1_{2,j} \rightarrow ⑦$$

In eq$^n$ ⑦ put $j=1$ & $j=2$

$$\left.\begin{aligned} u^2_{1,0} - 4u^2_{11} + u^2_{12} &= -u^1_{01} - u^1_{21} \\ u^2_{11} - 4u^2_{12} + u^2_{13} &= -u^1_{02} - u^1_{2,2} \end{aligned}\right\} ⑧$$

Substitute values in eq$^n$ ⑧

$$0 - 4u^2_{11} + u^2_{12} = -0 - 1/3$$

$$u^2_{11} - 4u^2_{12} + 1 = 0 - 1/3$$

$$u^2_{11} = \frac{8}{45} \qquad u^2_{12} = 17/45$$

$$= 0.1778 \qquad = 0.3778$$

In eq$^n$ ② $r=0,\ i=2$

$$u^2_{2,j-1} - 4u^2_{2j} + u^2_{2,j+1} = -u^1_{1,j} - u^1_{3,j} \longrightarrow ⑨$$

In eq$^n$ ⑨ put $j=1,\ j=2$

$$\left.\begin{aligned} u^2_{2,0} - 4u^2_{2,1} + u^2_{2,2} &= -u^1_{1,1} - u^1_{3,1} \\ u^2_{2,1} - 4u^2_{2,2} + u^2_{2,3} &= -u^1_{1,2} - u^1_{3,2} \end{aligned}\right\} \rightarrow ⑩$$

Substitute values in ⑩

$$0 - 4u^2_{2,1} + u^2_{2,2} = \cancel{0} - \cancel{0}\ -1/3 - 0$$

$$u^2_{2,1} - 4u^2_{2,2} + 1 = -1/3 - 0$$

$$\cancel{u_{2,1} = 2/17} \qquad \cancel{u_{2,2} = 8/17} \qquad u^2_{2,1} = 8/45, \quad u^2_{2,2} = 17/45$$

## Bairtow method

$$P_n(x) = a_0x^n + a_1x^{n-1} + a_2x^{n-2} + \dots + a_n$$

$$x^2 + px + q$$

$$\left.\begin{aligned} p_{k+1} &= p_k + \Delta p_k \\ q_{k+1} &= q_k + \Delta q_k \end{aligned}\right\} k = 0, 1, 2, \dots$$

$$\Delta p = -\left[\frac{b_n c_{n-3} - b_{n-1} c_{n-2}}{c_{n-2} c_{n-2} - c_{n-3} c_{n-1} + c_{n-3} b_{n-1}}\right] \begin{matrix} \to -3 \\ \to -4 \end{matrix}$$

$$\Delta q = -\left[\frac{b_n c_{n-1} - b_{n-1} b_{n-1} - b_n c_{n-2}}{c_{n-2} c_{n-2} - c_{n-3} c_{n-1} + c_{n-3} b_{n-1}}\right] \to -2$$

| | | | | | |
|---|---|---|---|---|---|
| $-p_0$ | $a_0$ | $a_1$ | $a_2$ | $a_3$ | $a_4$ |
| | | $-a_0p$ | $-pb_1$ | $-pb_2$ | $-pb_3$ |
| $-q_0$ | | | $-qb_0$ | $-qb_1$ | $-qb_2$ |
| | $a_0 = b_0$ | $b_1$ | $b_2$ | $b_3$ | $b_4$ |
| $-p_0$ | | $-pb_0$ | $-pc_1$ | $-pc_2$ | |
| $-q_0$ | | | $-qc_0$ | $-qc_1$ | |
| | $b_0 = c_0$ | $c_1$ | $c_2$ | $c_3$ | |

1) Using Bairtow method of obtain the quadratic factor of the eq$^n$ $x^4 - 3x^3 + 20x^2 + 44x + 54 = 0$ with $(p, q) = (2, 2)$

Sol: $n = 4$ $p_0 = 2$ $q_0 = 2$

$x^2 + px + q$

$p_{k+1} = p_k + \Delta p_k$

$q_{k+1} = q_k + \Delta q_k$

$$\Delta p = -\left[\frac{b_4 c_1 - b_3 c_2}{c_2^2 - c_1 c_3 + c_1 b_3}\right]$$

$$\Delta q = -\left[\frac{b_3 c_3 - b_3^2 - b_4 c_2}{c_2^2 - c_1 c_3 + c_1 b_3}\right]$$

$$\begin{array}{r|ccccc}
-2 & 1 & -3 & 20 & 44 & 54 \\
 & & -2 & 10 & -56 & 4 \\
-2 & & & -2 & 10 & -56 \\
\hline
-2 & 1 = b_0 & -5 = b_1 & 28 = b_2 & -2 = b_3 & 2 = b_4 \\
 & & -2 & 14 & -80 & \\
-2 & & & -2 & 14 & \\
\hline
 & 1 = c_0 & -7 = c_1 & 40 = c_2 & -68 = c_3 &
\end{array}$$

$$\Delta p = -\left[\frac{2(-7) - (-2)(40)}{(40)^2 - (-7)(-68) + (-7)(-2)}\right] = \cancel{-33}\ -0.0580$$

$$\Delta q = -\left[\frac{(-2)(-68) - (-2)^2 - (2)(40)}{(40)^2 - (-7)(-68) + (-7)(-2)}\right] = -0.0457$$

$\therefore$ ~~$\beta f$~~ $p_1 = p_0 + \Delta p_0$

$= 2 - 0.0580$

$= 1.9420$

$q_1 = q_0 + \Delta q_0$

$= 2 - 0.0457$

$= 1.9543$

| | | | | | |
|---|---|---|---|---|---|
| | 1 | -3 | 20 | 44 | $\cancel{84}$ 54 |
| -1.9420 | | -1.9420 | 9.5974 | -53.6829 | 0.0480 |
| -1.9543 | | | -1.9513 | +9.6582 | -54.0229 |
| | $1 = b_0$ | $-4.9420 = b_1$ | $27.6434 = b_2$ | $\cancel{63.3}$ $-0.0247 = b_3$ | $0.0251 = b_4$ |
| -1.9420 | | -1.9420 | 13.3687 | -75.8497 | |
| -1.9543 | | | -1.9543 | 13.4534 | |
| | $1 = c_0$ | $-6.8840 = c_1$ | $39.0575 = c_2$ | $-62.4210 = c_3$ | |

$$\Delta P_1 = -\left[\frac{(0.0251)(-6.8840) - (-0.0247)(39.0575)}{(39.0575)^2 - (-6.8840)(62.4210) + (-6.8840)(-0.0247)}\right]$$

$$= -0.0007$$

$$\Delta q_1 = -\left[\frac{(-0.0247)(-62.4210) - (-0.0247)^2 - (0.0251)(39.0575)}{(39.0575)^2 - (-6.8840)(-62.4210) + (-6.8840)(-0.0247)}\right]$$

$$= -0.0005$$

$P_2 = P_1 + \Delta P_1 = +1.9420 - 0.0007 = +1.94\cancel{07}13$

$q_2 = q_1 + \Delta q_1 = +1.9543 - 0.0005 = 1.9538$

∴ The required quadratic factor is $x^2 + (1.9413)x + 1.9538$

2) $x^4 - x^3 + 6x^2 + 5x + 10 = 0$ with $(p, q) = (1.14, 1.42)$

$n = 4$, $p_0 = 1.14$, $q_0 = 1.42$

$x^2 + px + q$

$$\Delta P = -\left[\frac{b_4 c_1 - b_3 c_2}{c_2^2 - c_1 c_3 + c_1 b_3}\right] \qquad \Delta q = -\left[\frac{b_3 c_3 - b_3^2 - b_4 c_2}{c_2^2 - c_1 c_3 + c_1 b_3}\right]$$

$P_{k+1} = P_k + \Delta P_k$

$q_{k+1} = q_k + \Delta q_k$

| | | | | | |
|---|---|---|---|---|---|
| | 1 | -1 | 6 | 5 | 10 |
| -1.14 | | -1.14 | 2.4396 | -8.0023 | -0.0416 |
| -1.42 | | | -1.42 | 3.0388 | -9.9678 |
| | 1 | -2.14 | 7.0196 | 0.0365 | -0.0094 |
| -1.14 | | -1.14 | 3.7392 | -10.6462 | |
| -1.42 | | | -1.42 | 4.6576 | |
| | 1 | -3.28 | 9.3388 | -5.9521 | |

$$\Delta P = -\left[\frac{(-0.0094)(-3.28) - (0.0365)(9.3388)}{(9.3388)^2 - (-3.28)(-5.9521) + (-3.28)(0.0365)}\right]$$

$= +0.0046$

$$\Delta q = -\left[\frac{(0.0365)(-5.9521) - \cancel{(9.3388)^2} - (0.0365)^2 - (-0.0094)(9.3388)}{(9.3388)^2 - (-3.28)(-5.9521) + (-3.28)(0.0365)}\right]$$

$= 0.0019$

$P_1 = 1.1446$, $q_1 = 1.4219$

| | | | | | |
|---|---|---|---|---|---|
| | 1 | $-1$ | 6 | 5 | 10 |
| $-1.1446$ | | $-1.1446$ | $2.1655$ | | |
| $-1.4219$ | | | $1.4219$ | | |
| | 1 | $-2.1446$ | $7.5874$ | | |

$\Delta p = -0.00003$

$\Delta q = 0.00003$

$p_2 = 1.1446$

$q_2 = 1.4219$

## Graffe's square root method:

$$x^4 + a_1 x^3 + a_2 x^2 + a_3 x + a_4 = 0$$

| $m$ | $2^m$ | | | | | |
|---|---|---|---|---|---|---|
| 0 | 1 | 1 | $a_1$ | $a_2$ | $a_3$ | $a_4$ |
| | | 1 | $a_1^2$ | $a_2^2$ | $a_3^2$ | $a_4^2$ |
| | | | $-2a_2(1)$ | $-2a_1a_3$ | $-2a_2a_4$ | |
| 1 | $2^1$ | 1 | $b_1$ | $b_2$ | $b_3$ | $b_4$ |
| | | 1 | $b_1^2$ | $b_2^2$ | $b_3^2$ | $b_4^2$ |
| | | | $-2b_2(1)$ | $-2b_1b_3$ | $-2b_2b_4$ | |
| 1 | $2^2$ | 1 | $c_1$ | $c_2$ | $c_3$ | $c_4$ |

| | $\alpha_1$ | $\alpha_2$ | $\alpha_3$ | $\alpha_4$ |
|---|---|---|---|---|
| 1st | $\left(\frac{b_1}{1}\right)^{1/2}$ | $\left(\frac{b_2}{b_1}\right)^{1/2}$ | $\left(\frac{b_3}{b_2}\right)^{1/2}$ | $\left(\frac{b_4}{b_3}\right)^{1/2}$ |
| 2nd | $\left(\frac{c_1}{1}\right)^{1/4}$ | $\left(\frac{c_2}{c_1}\right)^{1/4}$ | $\left(\frac{c_3}{c_2}\right)^{1/4}$ | $\left(\frac{c_4}{c_3}\right)^{1/4}$ |
| 3rd | $\left(\frac{d_1}{1}\right)^{1/8}$ | $\left(\frac{d_2}{d_1}\right)^{1/8}$ | $\left(\frac{d_3}{d_2}\right)^{1/8}$ | $\left(\frac{d_4}{d_3}\right)^{1/8}$ |
| 4th | $\left(\frac{e_1}{1}\right)^{1/16}$ | $\left(\frac{e_2}{e_1}\right)^{1/16}$ | $\left(\frac{e_3}{e_2}\right)^{1/16}$ | $\left(\frac{e_4}{e_3}\right)^{1/16}$ |

Find all the roots of the polynomial $x^3-6x^2+11x-6=0$ using Graffe's square root method.

$x^3-6x^2+11x-6=0$

| $m$ | $2^m$ | | | | |
|---|---|---|---|---|---|
| 0 | $2^0=1$ | 1 | -6 | 11 | -6 |
| | | 1 | 36 | 121 | 36 |
| | | | -22 | -72 | |
| 1 | 2 | 1 | 14 | 49 | 36 |
| | | 1 | 196 | 2401 | 1296 |
| | | | -98 | -1008 | |
| 2 | 4 | 1 | 98 | 1393 | 1296 |
| | | 1 | 9604 | 1940449 | 1679616 |
| | | | -2786 | -254016 | |
| 3 | 8 | 1 | 6818 | 1686433 | 1679616 |
| | | 1 | 4648524 | $2.84\times10^{12}$ | $2.8$[illegible]$\times10^{12}$ |
| | | | -3372866 | $-2.29032\times10^{10}$ | |
| 4 | 16 | 1 | 43112258 | $2.8211\times10^{12}$ | $2.8211\times10^{12}$ |

The approximate roots are

| $\alpha_1$ | $\alpha_2$ | $\alpha_3$ |
|---|---|---|
| 3.7417 | 1.8708 | 0.8571 |
| 3.1463 | 1.9417 | 0.9821 |
| 3.0144 | 1.9914 | 0.9995 |
| 3.0003 | 1.9998 | 1 |

$\therefore$ The exact roots are 3, 2, 1

Find all the roots of the polynomial $x^6-4.8x^4+3.33x^2-0.0052044$

put $x^2=t$ $\quad p(t)=t^3-4.8t^2+3.3t-0.005=0$

| $m$ | $2^m$ | | | | |
|---|---|---|---|---|---|
| 0 | 1 | 1 | $-4.8$ | $3.3$ | $-0.005$ |
| | | 1 | $23.04$ | $10.89$ | $2.8\times10^{-3}$ |
| | | | $-6.6$ | $-0.0480$ | |
| | | 1 | $16.44$ | $\cancel{10.8420}$ | $2.5\times10^{-3}$ |
| 1 | 2 | | | $10.41$ | |
| | | 1 | $270.2736$ | $\cancel{117.549}$ | $6.25\times10^{-6}$ |
| | | | $-20.82$ | $108.3681$ | |
| | | | $\cancel{-235.0980}$ | $\cancel{-3.37842\times10^{-3}}$ | |
| | | | $\cancel{-21.684}$ | $-0.0822$ | |
| 2 | 4 | 1 | $\cancel{248.5896}$ | $\cancel{117.4668}$ | $6.25\times10^{-6}$ |
| | | | $249.4536$ | $108.2859$ | |
| | | 1 | $\cancel{6176.789}$ | $\cancel{13798.41}$ | |
| | | | $62227.0986$ | $11725.8361$ | $0.03906\times10^{-10}$ |
| | | | $-2.165718$ | $-0.0031$ | |
| 3 | 8 | 1 | $62010.5268$ | $11725.833$ | $0.3906\times10^{-10}$ |

| $m$ | $\alpha_1$ | $\alpha_2$ | $\alpha_3$ |
|---|---|---|---|
| 1 | 4.0546 | 0.7957 | 0.0155 |
| 2 | 3.9741 | 0.8116 | 0.0154 |
| ~~3~~ 3 | 3.9724 | 0.8120 | 0.0154 |

The exact roots of polynomial $p(t)$ are

3.9724 $\quad$ 0.812 $\quad$ 0.0154

$\pm 1.9933$ $\quad \pm 0.9011$ $\quad \pm 0.124$